

# A consistent iterative scheme for 2D and 3D cohesive crack analysis in XFEM



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## ARTICLE INFO

### Article history:

Received 29 November 2012

Accepted 28 January 2014

### Keywords:

Fracture mechanics  
Cohesive crack model  
XFEM

## ABSTRACT

This paper presents a consistent algorithm for cohesive crack modeling with the extended finite element method (XFEM). A new formulation for the transformation from global to local quantities for the cohesive law is presented. This formulation works as a generic shell around any cohesive law so that the cohesive law needs only to be implemented in two dimensions while it can be used in both two and three dimensional analysis. Particular attention is paid to consistent linearization of the contribution from the cohesive tractions. The proposed formulation is tested on three relevant examples of three-dimensional analysis.

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## 1. Introduction

Engineering structures have to be analyzed with respect to their failure possibilities in order to ensure their safety. Tensile stresses are particularly dangerous to many materials because they may cause decohesion of the material due to a gradual loss of micro-structural load-bearing capacity. Eventually a crack appears where displacements are not continuous on the structural level of observation. The analysis of fracture problems is usually carried out with either damage mechanics or fracture mechanics. In linear elastic fracture mechanics, the failure of a material is characterized by the growth of a discrete crack through an otherwise intact medium. In most materials, however, the real crack is preceded by the so-called fracture process zone where tensile stresses decrease. The materials where the fracture process zone is not negligibly small are usually called quasi-brittle materials. An accurate description of the failure of such materials requires that the fracture process zone is taken into account in the analysis. A popular way to do this is with the so-called cohesive crack model [1–3]. In the cohesive crack model it is assumed that near the crack tip tractions act from one crack face to the other. These tractions decrease as the crack opening increases and eventually the crack becomes traction-free.

Computational modelling of the crack process was usually carried out by finite element method (FEM) e.g. [4,5]. The element-free Galerkin method (EFGM) was also used in

application of crack growth analysis (e.g. [6–9]), or the method based on coupled FEM–EFGM (e.g. [10,11]). Nowadays the methods based on partition of unity method (PUM) are seen as the most accurate technology for crack propagation analysis. The extended finite element method (XFEM) is the most widely used [12–15], but extended EFGM can also applied [16,10].

The main advantage of computer methods based on PUM is the fact that arbitrary functions can be incorporated in the approximation. In the case of fracture analysis, the Heaviside step function is used to incorporate the discontinuity over the crack surface. An arbitrary number of additional functions can be used in the formulation, which is why additional enrichment functions are sometimes used near the crack tip [12,17,11,18–21]. These additional enrichment functions usually refer to an analytical solution for the displacement field around the crack tip. In the cohesive crack model there is no singularity at the crack tip and therefore the crack tip enrichment with singular functions is not necessary. With cohesive methods, a stress criterion is used for crack initiation, while the fracture toughness plays a role in the propagation through the cohesive law. This combined approach with fracture stress and toughness as material parameters is typical for quasi-brittle materials.

In the cohesive crack model, cohesive forces act along the crack in a fracture process zone. At every point on the crack surface, there is a crack opening vector that can be decomposed into the normal  $w_n$  and sliding  $w_s$  parts, where normal and sliding directions refer to the crack orientation. In most analyses, the cohesive forces act in the normal direction and are described by a constitutive cohesion law in terms  $w_n$ . When mixed-mode crack-ing is analyzed, additional forces that resist the sliding deformation

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between the crack faces can come into play. Then, an additional constitutive law is set for these sliding forces which depend on both crack opening parts  $w_n$  and  $w_s$  as in [13]. Sometimes the total crack opening is used to construct the cohesion traction-separation law (e.g. [8,18,22]). In this paper, the general assumptions are made that normal and sliding cohesion forces are taken into account and that both depend on  $w_n$  and  $w_s$ . Based on these assumptions general relations are derived.

The cohesive crack propagation analysis is a nonlinear problem. Firstly, the nonlinearity of the process is connected with the propagation of the discontinuity. It results in a decrease in the stiffness of the structure. Secondly, the cohesion forces along the crack depend on the actual solution. In order to find the solution of this nonlinear problem, a Newton–Raphson scheme must be applied, which means that the state in which internal forces are in equilibrium with external forces is found iteratively. This iterative procedure is most effective when a tangential stiffness matrix is calculated. Therefore, a procedure is needed to linearize the relation between cohesive forces and the crack opening vector.

The tangential stiffness matrix includes a second order tangent cohesive stiffness matrix  $\mathbf{C}$  from the cohesion traction-separation law and the shape functions jump. This kind of matrix was used for example in [13–15,23] but it was evaluated for standard XFEM formulation with the Heaviside step function and basic cohesion constitutive law. In [18] the tangent matrix was evaluated for local coordinate system connected with the crack line. A more general idea was proposed in [11,16] where the matrix was developed by differentiating the cohesion force vector with respect to displacement jump vector in the global coordinate system which was applied to mode I cohesive cracks. In this paper the relations from [11] are generalized to cracks in three dimensions (3D) with general cohesive constitutive laws. The tangent matrix is evaluated in global coordinate system, while the cohesive law is evaluated in a two-dimensional (2D) local coordinate frame aligned with the crack surface.

In the past decade there are plenty of works concerning fracture simulation by XFEM. Some example analysis of three-dimensional crack analysis can be found in [24–26] while recent developments connected with cohesive crack models have been presented in [27–33]. A review of XFEM has been given in [9].

The paper is organized as follows: In Section 2 a mathematical model for the quasi-static cohesive cracking problem is developed. In Section 3 a new approach is presented for calculating and applying cohesion forces in cracks in 2D and 3D. The presented algorithm is verified by examples presented in Section 4. In Section 5 conclusions are presented.

## 2. Mathematical model

The computational model in the paper is derived for full 3D case. However, the relations that are derived below can however be simplified to the 2D case in a plane stress or plane strain state. It is assumed that:

1. strains and displacement are small,
2. the relation between stress and strain is linear,
3. there is a crack in the body denoted by  $S_d$ . In 3D the crack is a curved plane, in 2D the crack is a curved line,
4. along  $S_d$  the displacement field is discontinuous,
5. there is a fracture process zone in the crack that is modeled by cohesion forces  $\mathbf{t}_c$ .

In the paper Voigt notation is used for stress  $\boldsymbol{\sigma}$ , strain  $\boldsymbol{\varepsilon}$  and matrix  $\mathbf{m}$  representing the normal to the surface of the analyzed body.

The analysis begins with the standard equilibrium equation (momentum balance) that is valid at each point of the considered isotropic solid and for each moment of time:

$$\mathbf{L}^T \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \quad \text{in } V, \quad (1)$$

where  $\mathbf{b}$  is the body force vector and  $\mathbf{L}$  a matrix of differential operators.

The equation is completed by the standard boundary conditions and equilibrium conditions for the crack plane:

$$\begin{aligned} \mathbf{m}^T \boldsymbol{\sigma} &= \mathbf{t} \quad \text{on } S_\sigma, \\ \mathbf{u} &= \hat{\mathbf{u}} \quad \text{on } S_u, \\ \mathbf{m}_d^T \boldsymbol{\sigma} &= \mathbf{t}_c \quad \text{on } S_d, \end{aligned} \quad (2)$$

where  $\mathbf{t}$  is the vector of external tractions,  $\mathbf{u}$  is the displacement vector,  $\hat{\mathbf{u}}$  is the vector of prescribed displacements,  $\mathbf{m}_d$  represents the normal to the crack surface in Voigt notation, and  $\mathbf{t}_c$  is the vector of cohesive forces along the crack surface.

The local mathematical model from Eqs. (1) and (2) is reworked into a global weak formulation using the weighted residual approach, where a test function  $\mathbf{v}_u$  is used. The test function  $\mathbf{v}_u$  and displacement field  $\mathbf{u}$  are discontinuous in  $S_d$ . Therefore Dirac's delta appears in  $S_d$  while differentiating the fields

$$\mathbf{L}\mathbf{u} = \begin{cases} \mathbf{L}\mathbf{u} & \text{if } \mathbf{x} \notin S_d, \\ \delta_{S_d} \mathbf{m}_d \llbracket \mathbf{u} \rrbracket & \text{if } \mathbf{x} \in S_d. \end{cases} \quad (3)$$

where:

- $\delta_{S_d}$  – Dirac's delta along crack  $S_d$ ,
- $\mathbf{m}_d$  – normal to  $S_d$  in Voigt notation,
- $\llbracket \cdot \rrbracket$  – function discontinuity operator.

Supposing that  $\mathbf{n}_d$  is a normal vector to  $S_d$  in  $\mathbf{x} \in S_d$ , the discontinuity of the test function (or any other quantity) is

$$\llbracket \mathbf{v}_u \rrbracket(\mathbf{x}) = \lim_{\lambda \rightarrow 0} \mathbf{v}_u(\mathbf{x} + \lambda \mathbf{n}_d) - \lim_{\lambda \rightarrow 0} \mathbf{v}_u(\mathbf{x} - \lambda \mathbf{n}_d) = \mathbf{v}_u^+(\mathbf{x}) - \mathbf{v}_u^-(\mathbf{x}). \quad (4)$$

In that case the equilibrium equation in weak form at time  $t + \Delta t$  is

$$\begin{aligned} \int_V (\mathbf{L}\mathbf{v}_u)^T \boldsymbol{\sigma}^{t+\Delta t} dV + \int_{S_d} \delta_{S_d} \llbracket \mathbf{v}_u \rrbracket^T \mathbf{m}_d^T \boldsymbol{\sigma}^{t+\Delta t} dV - \int_V \mathbf{v}_u^T \mathbf{b}^{t+\Delta t} dV \\ - \int_{S_d} \mathbf{v}_u^T \mathbf{t}_c^{t+\Delta t} dS = 0, \end{aligned} \quad (5)$$

where  $t$  is the pseudo-time connected with the external forces that is usually used in quasi-static analysis,  $\delta_{S_d}$  is Dirac's delta along crack  $S_d$ .

With the definition of Dirac's delta, the weak formulation can be rewritten as:

$$\int_V (\mathbf{L}\mathbf{v}_u)^T \boldsymbol{\sigma}^{t+\Delta t} dV + \int_{S_d} \llbracket \mathbf{v}_u \rrbracket^T \mathbf{t}_c^{t+\Delta t} dS - \int_V \mathbf{v}_u^T \mathbf{b}^{t+\Delta t} dV - \int_{S_d} \mathbf{v}_u^T \mathbf{t}_c^{t+\Delta t} dS = 0. \quad (6)$$

where  $\mathbf{t}_c$  depends on the discontinuity in the displacements,  $\llbracket \mathbf{u} \rrbracket$ , that is referred to in this paper as the crack opening vector.

For the solution of the nonlinear system of equations the standard incremental-iterative procedure is applied. It is supposed that all quantities are known for time  $t$ . The displacement in time  $t + \Delta t$  is then defined as the sum of the displacement at time  $t$  and an unknown increment

$$\mathbf{u}^{t+\Delta t} = \mathbf{u}^t + \Delta \mathbf{u}. \quad (7)$$

The displacement increment  $\Delta \mathbf{u}$  is computed iteratively and it can be defined as the limit of a sequence, where an element of the sequence is then calculated by adding a corrective increment  $d\mathbf{u}$  to the previous element of the sequence:

$$\Delta \mathbf{u} = \lim_{i \rightarrow \infty} \Delta \mathbf{u}_i, \quad \Delta \mathbf{u}_{i+1} = \Delta \mathbf{u}_i + d\mathbf{u}. \quad (8)$$

For all other quantities the same procedure is performed, including the crack opening vector  $\llbracket \mathbf{u} \rrbracket$ .

In each iteration of the Newton–Raphson procedure,  $\mathbf{du}$  is computed by solving the linearized system of equations. For linear stress–strain law and non-linear cohesive law, this takes the form of:

$$\int_V (\mathbf{L}\mathbf{v}_u)^T \mathbf{E} \mathbf{L} \mathbf{du} dV + \int_{S_d} \llbracket \mathbf{v}_u \rrbracket^T \mathbf{C}_i d\llbracket \mathbf{u} \rrbracket dS = W_{\text{ext}}^{t+\Delta t} - W_{\text{int},i}^{t+\Delta t}, \quad (9)$$

where

$$W_{\text{int},i}^{t+\Delta t} = \int_V (\mathbf{L}\mathbf{v}_u)^T \boldsymbol{\sigma}_i^{t+\Delta t} dV + \int_{S_d} \llbracket \mathbf{v}_u \rrbracket^T \mathbf{t}_i^{t+\Delta t} dS, \quad (10)$$

$$W_{\text{ext}}^{t+\Delta t} = \int_V \mathbf{v}_u^T \mathbf{b}^{t+\Delta t} dV + \int_{S_\sigma} \mathbf{v}_u^T \mathbf{t}^{t+\Delta t} dS, \quad (11)$$

and  $\mathbf{E}$  is the material stiffness matrix from Hooke's law and  $\mathbf{C}$  is the tangent matrix for the relation between the cohesive traction and the crack opening:

$$\mathbf{C} = \frac{\partial \mathbf{t}_c}{\partial \llbracket \mathbf{u} \rrbracket}. \quad (12)$$

For the discretization of the field equation with XFEM, we refer to [13]. The focus of the current paper is on the evaluation of cohesive tractions  $\mathbf{t}_c$  and the associated tangent  $\mathbf{C}$ .

### 3. The cohesion vector and matrix

The cohesive traction vector  $\mathbf{t}_c$  and its related tangent stiffness matrix  $\mathbf{C}$  generally depend on constitutive laws that are described in relation to normal and sliding components of the crack opening vector. Therefore at every integration point in  $S_d$  a local coordinate system has to be constructed. The cohesive tractions and their derivatives are then evaluated in the local coordinate system and transformed back to the global coordinate system. The main novelty of this paper lies in the way this local coordinate frame is defined. In this section, a consistent formulation is derived for which the constitutive law can always be evaluated in 2D, even when global jump and tractions are defined in 3D. In Section 3.1 the transformation from global to local frame and vice versa is introduced. It is emphasized that the procedure in Section 3.1 is generic in the sense that it can be applied to any constitutive law that relates cohesive tractions to the displacement jump. To achieve this, firstly the local coordinate system and the displacement jump in local coordinate frame are defined. Then the transformations of the traction vector and tangent matrix from local to global frame are described. In Sections 3.2 and 3.3 the two cohesive laws that are used for the numerical examples are presented to complete the formulation.

#### 3.1. Local to global transformations

The local coordinate system at a point that belongs to  $S_d$  is defined by the unit vector normal to crack at that point  $\mathbf{n}_d$  and the unit vector parallel to the sliding part of crack opening vector  $\mathbf{s}_d$ . The local coordinate system always has a 2D character, for both 2D and 3D analysis in the sense that the crack opening and traction in the third perpendicular direction is by definition equal to zero. As a consequence, the formulation of the cohesive law becomes independent of whether it is used for 3D or 2D analysis. In 2D analysis, the orientation of  $\mathbf{s}_d$  is fixed as the tangent vector to the crack line. But in general, it depends on the actual solution, so it may happen that the orientation of  $\mathbf{s}_d$  changes during the Newton–Raphson procedure.

The decomposition of crack opening vector into normal and sliding part was firstly proposed for explicit 3D cohesive crack analysis by Ortiz and Pandolfi in [34]. The idea of decomposing the crack opening vector this way has subsequently been applied successfully in a number of other publications, for example in [29,35–38]. In this paper, the idea of vector decomposition is pursued further to efficiently obtain the tangent stiffness matrix for the Newton–Raphson iterative procedure in implicit analysis.

The crack opening vector  $\llbracket \mathbf{u} \rrbracket$  can be decomposed into a normal part and a sliding part, where the normal part of the vector is obtained with:

$$\mathbf{w}_n = \mathbf{n}_d^T \llbracket \mathbf{u} \rrbracket. \quad (13)$$

Then the normal crack opening vector is given as:

$$\mathbf{w}_n = \mathbf{n}_d \mathbf{w}_n = \mathbf{n}_d \mathbf{n}_d^T \llbracket \mathbf{u} \rrbracket, \quad (14)$$

which means that the vector  $\mathbf{w}_n$  is the projection of  $\llbracket \mathbf{u} \rrbracket$  in the direction of  $\mathbf{n}_d$ .

The sliding crack opening vector is calculated as the difference of  $\llbracket \mathbf{u} \rrbracket$  and  $\mathbf{w}_n$ :

$$\mathbf{w}_s = \llbracket \mathbf{u} \rrbracket - \mathbf{w}_n, \quad (15)$$

$$\mathbf{w}_s = \llbracket \mathbf{u} \rrbracket - \mathbf{n}_d \mathbf{n}_d^T \llbracket \mathbf{u} \rrbracket, \quad (16)$$

$$\mathbf{w}_s = (\mathbf{I} - \mathbf{n}_d \mathbf{n}_d^T) \llbracket \mathbf{u} \rrbracket. \quad (17)$$

This means that  $\mathbf{w}_s$  is the projection of  $\llbracket \mathbf{u} \rrbracket$  onto the crack surface  $S_d$ . Subsequently, the unit vector in the sliding direction is given by

$$\mathbf{s}_d = \frac{\mathbf{w}_s}{\|\mathbf{w}_s\|}, \quad (18)$$

The magnitude of the sliding part of crack opening vector then becomes

$$w_s = \mathbf{s}_d^T \llbracket \mathbf{u} \rrbracket. \quad (19)$$

The vectors  $\mathbf{n}_d$  and  $\mathbf{s}_d$  define a 2D local coordinate frame in the plane that is spanned by the normal vector to the crack surface and the crack opening vector. The normal vector to the crack  $\mathbf{n}_d$  is connected with the crack geometry and therefore constant in time (although it may vary in space). The sliding vector  $\mathbf{s}_d$  depends on the crack opening vector  $\llbracket \mathbf{u} \rrbracket$  and may therefore change in time. It should be noted that it is possible at some points along the crack that the opening  $\llbracket \mathbf{u} \rrbracket$  is perpendicular to the crack surface, i.e. in which  $\mathbf{w}_s = 0$ . In such a situation the sliding direction  $\mathbf{s}_d$  cannot be determined. That is why there is a necessity to distinguish between the following two cases:

1. With crack sliding. The sliding part of crack opening vector  $\llbracket \mathbf{u} \rrbracket$  is significant. It can be stated that the crack sliding part is large enough to be taken into account when the following condition is met:

$$1 - \frac{w_n}{\|\llbracket \mathbf{u} \rrbracket\|} > \epsilon, \quad (20)$$

where  $\epsilon$  is a small number that has to be chosen in such a way as to guarantee numerical stability.

2. Without crack sliding. The crack sliding part is negligibly small, which can be written with the condition:

$$1 - \frac{w_n}{\|\llbracket \mathbf{u} \rrbracket\|} \leq \epsilon. \quad (21)$$

In the remainder of this section, general relations are first discussed for the case with crack sliding. Then the general relations are simplified for the case without crack sliding.

A basic assumption for the cohesive law is that the sliding traction acts in the direction of the sliding part of the crack opening. In this case,  $\mathbf{t}_c$  can be decomposed into normal and sliding parts with

$$\mathbf{t}_c = \mathbf{n}_d t_n + \mathbf{s}_d t_s. \quad (22)$$

Examples of constitutive laws for which this holds can be found in [18,39–41]. However, cohesive laws that include dry friction effects do not follow this assumption. The current formulation is not valid for such cohesive laws in 3D.

It is supposed that the normal  $t_n$  and sliding  $t_s$  cohesion forces are defined by a constitutive law in relation to normal  $w_n$  and sliding  $w_s$  parts of crack opening vector

$$\mathbf{t}_{ns} = \mathbf{t}_{ns}(w_n, w_s), \quad (23)$$

where

$$\mathbf{t}_{ns} = \begin{bmatrix} t_n \\ t_s \end{bmatrix}. \quad (24)$$

Usually, the constitutive laws use their history parameters to take into account the difference between loading and unloading situations.

In Eq. (12) the constitutive cohesion stiffness matrix is defined by differentiating the cohesion vector with respect to the crack opening vector. In order to evaluate the matrix Eqs. (22) and (23) are taken into account

$$\mathbf{C} = \frac{\partial \mathbf{t}_c}{\partial [\mathbf{u}]} = \mathbf{n}_d \frac{\partial t_n}{\partial [\mathbf{u}]} + \mathbf{s}_d \frac{\partial t_s}{\partial [\mathbf{u}]} + \frac{\partial \mathbf{s}_d}{\partial [\mathbf{u}]} t_s. \quad (25)$$

The last element in Eq. (25) comes from the fact that the sliding direction is a function of the crack opening vector  $[\mathbf{u}]$ .

From Eqs. (13) and (19) the following relations are obtained:

$$\frac{\partial w_n}{\partial [\mathbf{u}]} = \mathbf{n}_d^T, \quad \frac{\partial w_s}{\partial [\mathbf{u}]} = \mathbf{s}_d^T, \quad (26)$$

With these relations, the partial derivatives from Eq. (25) can be written as follows:

$$\frac{\partial t_n}{\partial [\mathbf{u}]} = \frac{\partial t_n}{\partial w_n} \frac{\partial w_n}{\partial [\mathbf{u}]} + \frac{\partial t_n}{\partial w_s} \frac{\partial w_s}{\partial [\mathbf{u}]} = \mathbf{n}_d^T \frac{\partial t_n}{\partial w_n} + \mathbf{s}_d^T \frac{\partial t_n}{\partial w_s} \quad (27)$$

$$\frac{\partial t_s}{\partial [\mathbf{u}]} = \frac{\partial t_s}{\partial w_n} \frac{\partial w_n}{\partial [\mathbf{u}]} + \frac{\partial t_s}{\partial w_s} \frac{\partial w_s}{\partial [\mathbf{u}]} = \mathbf{n}_d^T \frac{\partial t_s}{\partial w_n} + \mathbf{s}_d^T \frac{\partial t_s}{\partial w_s} \quad (28)$$

$$\frac{\partial \mathbf{s}_d}{\partial [\mathbf{u}]} = \frac{\partial (\mathbf{w}_s / w_s)}{\partial [\mathbf{u}]} = \frac{w_s (\mathbf{I} - \mathbf{n}_d \mathbf{n}_d^T) - \mathbf{w}_s \mathbf{s}_d^T}{w_s^2} = \frac{\mathbf{I} - \mathbf{n}_d \mathbf{n}_d^T - \mathbf{s}_d \mathbf{s}_d^T}{w_s}. \quad (29)$$

After substitution of these expressions into (25), the matrix  $\mathbf{C}$  takes the form:

$$\mathbf{C} = \mathbf{n}_d \mathbf{n}_d^T \frac{\partial t_n}{\partial w_n} + \mathbf{n}_d \mathbf{s}_d^T \frac{\partial t_n}{\partial w_s} + \mathbf{s}_d \mathbf{n}_d^T \frac{\partial t_s}{\partial w_n} + \mathbf{s}_d \mathbf{s}_d^T \frac{\partial t_s}{\partial w_s} + \mathbf{S}, \quad (30)$$

where the matrix  $\mathbf{S}$  is defined as:

$$\mathbf{S} = (\mathbf{I} - \mathbf{n}_d \mathbf{n}_d^T - \mathbf{s}_d \mathbf{s}_d^T) \frac{\mathbf{t}_s}{w_s}. \quad (31)$$

It should be noted that in the 2D case the following relation is satisfied:

$$\mathbf{n}_d \mathbf{n}_d^T + \mathbf{s}_d \mathbf{s}_d^T = \mathbf{I}, \quad (32)$$

while in the 3D case a similar relation holds:

$$\mathbf{n}_d \mathbf{n}_d^T + \mathbf{s}_d \mathbf{s}_d^T + \mathbf{r}_d \mathbf{r}_d^T = \mathbf{I}. \quad (33)$$

where vector  $\mathbf{r}_d$  is defined as a cross product of  $\mathbf{n}_d$  and  $\mathbf{s}_d$  vectors

$$\mathbf{r}_d = \mathbf{n}_d \times \mathbf{s}_d. \quad (34)$$

With Eqs. (31) and (33) the matrix  $\mathbf{S}$  can also be written as:

$$\mathbf{S} = \begin{cases} \mathbf{0} & \text{in 2D,} \\ (\mathbf{r}_d \mathbf{r}_d^T) \frac{t_s}{w_s} & \text{in 3D.} \end{cases} \quad (35)$$

That  $\mathbf{S}$  equals zero for the 2D case is related to the fact that the  $\mathbf{s}_d$  vector is constant there.

Eqs. (22) and (30) can be expressed with the help of transformation matrix defined as follows:

$$\mathbf{Q} = [\mathbf{n}_d \quad \mathbf{s}_d]. \quad (36)$$

Then Eq. (22) takes the form

$$\mathbf{t}_c = \mathbf{Q} \mathbf{t}_{ns}, \quad (37)$$

Finally, matrix  $\mathbf{C}$  can be written:

$$\mathbf{C} = \mathbf{Q} \mathbf{C}_{ns} \mathbf{Q}^T + \mathbf{S}, \quad (38)$$

where  $\mathbf{C}_{ns}$  is tangent cohesion stiffness matrix in crack local coordinate system

$$\mathbf{C}_{ns} = \begin{bmatrix} \frac{\partial t_n}{\partial w_n} & \frac{\partial t_n}{\partial w_s} \\ \frac{\partial t_s}{\partial w_n} & \frac{\partial t_s}{\partial w_s} \end{bmatrix}. \quad (39)$$

The above definitions are derived under the assumption that the condition in Eq. (20) is met. In case the sliding part of the crack opening is negligible, the above relations can be simplified to eliminate the undetermined  $\mathbf{s}_d$ . The vector of cohesion forces from (22) changes to

$$\mathbf{t}_c = \mathbf{n}_d t_n, \quad (40)$$

and the cohesion stiffness matrix from (30) simplifies to

$$\mathbf{C} = \mathbf{n}_d \mathbf{n}_d^T \frac{\partial t_n}{\partial w_n}. \quad (41)$$

The constitutive laws from (23) do not depend on the sliding crack opening vector orientation. The sign of  $w_s$  does not matter (although in the current formulation it is always positive). The situation with the normal part of the crack opening is quite different. The constitutive laws are defined for  $w_n \geq 0$ . A negative value  $w_n$  would mean that two sides of the crack penetrate each other, which is physically unacceptable. In numeric simulation different techniques are used to prevent such situation (e.g. a penalty procedure, Kuhn-Tucker condition). In any case, it is crucial in the numerical analysis to calculate the normal part of crack opening with appropriate sign. This is particularly important in complex cases (e.g. with many cracks) where it is impossible to exclude the possibility of that crack closure. When the crack opening vector is calculated according to (4) and the normal part of the crack opening is calculated with (13), it is guaranteed that the  $w_n$  has the appropriate sign.

### 3.2. Turon cohesive law in local 2D coordinate frame

In the numerical examples in this paper, two different cohesive laws are applied. In this subsection and the following, the two laws are reviewed. Firstly, the mixed-mode cohesive law by Turon et al. [42] is discussed. This cohesive law has been developed for the simulation of delamination in composite laminates using the phenomenological relation that Benzeggagh and Kenane [43] proposed for the fracture toughness under mixed-mode conditions.

The traction and displacement jump are defined in the local frame that is aligned with crack surface, with the first component normal to that surface. Similar to Eq. (24), we write for the displacement jump in local coordinate frame:

$$\mathbf{w}_{ns} = \begin{bmatrix} w_n \\ w_s \end{bmatrix}. \quad (42)$$

The traction is computed from the displacement jump with isotropic damage and prevention of interpenetration as:

$$\mathbf{t}_{ns} = K[\mathbf{I} - d\mathbf{P}]\mathbf{w}_{ns}, \quad (43)$$

where  $K$  is the initial “dummy” stiffness of the interface,  $d$  is the damage variable, and the matrix  $\mathbf{P}$  is included to cancel the influence of damage in the normal direction in case of compression. This matrix is defined as:

$$\mathbf{P} = \begin{bmatrix} \frac{\langle w_n \rangle}{w_n} & 0 \\ 0 & 1 \end{bmatrix}, \quad (44)$$

using the operator  $\langle x \rangle = (x + |x|)/2$ .

The update of the damage variable is done with

$$d^{t+\Delta t} = \max(d^t, d_{\text{trial}}^{t+\Delta t}), \quad (45)$$

where

$$d_{\text{trial}}^{t+\Delta t} = \begin{cases} 0, & w_{\text{eq}} \leq w_{\text{eq}}^0 \\ \frac{w_{\text{eq}}^f (w_{\text{eq}} - w_{\text{eq}}^0)}{w_{\text{eq}} (w_{\text{eq}}^f - w_{\text{eq}}^0)}, & w_{\text{eq}}^0 < w_{\text{eq}} < w_{\text{eq}}^f \\ 1, & w_{\text{eq}} \geq w_{\text{eq}}^f \end{cases}. \quad (46)$$

Quantities on the right hand side Eq. (46) are computed from the displacement jump in local frame at time  $t + \Delta t$  (skipping the index for simplicity) as:

$$w_{\text{eq}} = \sqrt{\langle w_n \rangle^2 + w_s^2}, \quad (47)$$

$$w_{\text{eq}}^0 = \sqrt{(w_n^0)^2 + ((w_s^0)^2 - (w_n^0)^2)\alpha^\eta}, \quad (48)$$

$$w_{\text{eq}}^f = \frac{w_n^0 w_n^f + (w_s^0 w_s^f - w_n^0 w_n^f)\alpha^\eta}{w_{\text{eq}}^0}, \quad (49)$$

with

$$\alpha = \frac{w_s^2}{w_s^2 + \langle w_n \rangle^2}, \quad (50)$$

$$w_n^0 = \frac{F_t}{K}, \quad w_n^f = \frac{2G_{Ic}}{F_t}, \quad (51)$$

$$w_s^0 = \frac{F_{sh}}{K}, \quad w_s^f = \frac{2G_{IIc}}{F_{sh}}, \quad (52)$$

where  $F_t$  and  $F_{sh}$  are the tensile and shear strength,  $G_{Ic}$  and  $G_{IIc}$  are the mode I and mode II fracture toughness and  $\eta$  is the Benzeggagh–Kenane interaction parameter [43].

When damage is not changing, the gradient matrix of tractions with respect to the displacement jump is equal to the secant stiffness matrix. During damage growth, however, the consistent tangent matrix has a more complex expression. In general, the consistent tangent is written as:

$$\mathbf{C}_{ns} = \begin{cases} K[\mathbf{I} - d\mathbf{P}], & \dot{d} = 0 \\ K\left[\mathbf{I} - d\mathbf{P} - \mathbf{P} \mathbf{w}_{ns} \left(\frac{\partial d}{\partial \mathbf{w}_{ns}}\right)^T\right], & \dot{d} > 0 \end{cases} \quad (53)$$

with:

$$\frac{\partial d}{\partial \mathbf{w}_{ns}} = \frac{\partial d}{\partial w_{\text{eq}}} \frac{\partial w_{\text{eq}}}{\partial \mathbf{w}_{ns}} + \frac{\partial d}{\partial w_{\text{eq}}^0} \frac{\partial w_{\text{eq}}^0}{\partial \mathbf{w}_{ns}} + \frac{\partial d}{\partial w_{\text{eq}}^f} \frac{\partial w_{\text{eq}}^f}{\partial \mathbf{w}_{ns}} \quad (54)$$

Eq. (54) is expanded with

$$\frac{\partial d}{\partial w_{\text{eq}}} = \frac{w_{\text{eq}}^f w_{\text{eq}}^0}{w_{\text{eq}}^2 (w_{\text{eq}}^f - w_{\text{eq}}^0)} \quad (55)$$

$$\frac{\partial w_{\text{eq}}}{\partial \mathbf{w}_{ns}} = \frac{1}{w_{\text{eq}}} \mathbf{w}_{ns} \quad (56)$$

$$\frac{\partial d}{\partial w_{\text{eq}}^0} = -\frac{w_{\text{eq}}^f (w_{\text{eq}}^f - w_{\text{eq}}^0)}{w_{\text{eq}} (w_{\text{eq}}^f - w_{\text{eq}}^0)^2} \quad (57)$$

$$\frac{\partial w_{\text{eq}}^0}{\partial \mathbf{w}_{ns}} = \frac{\partial w_{\text{eq}}^0}{\partial \alpha^\eta} \frac{\partial \alpha^\eta}{\partial \alpha} \frac{\partial \alpha}{\partial \mathbf{w}_{ns}} \quad (58)$$

$$\frac{\partial d}{\partial w_{\text{eq}}^f} = -\frac{w_{\text{eq}}^0 (w_{\text{eq}} - w_{\text{eq}}^0)}{w_{\text{eq}} (w_{\text{eq}}^f - w_{\text{eq}}^0)^2} \quad (59)$$

$$\frac{\partial w_{\text{eq}}^f}{\partial \mathbf{w}_{ns}} = \frac{\partial w_{\text{eq}}^f}{\partial \alpha^\eta} \frac{\partial \alpha^\eta}{\partial \alpha} \frac{\partial \alpha}{\partial \mathbf{w}_{ns}} \quad (60)$$

and

$$\frac{\partial w_{\text{eq}}^0}{\partial \alpha^\eta} = \frac{(w_s^0)^2 - (w_n^0)^2}{2w_{\text{eq}}^0} \quad (61)$$

$$\frac{\partial w_{\text{eq}}^f}{\partial \alpha^\eta} = -\frac{w_{\text{eq}}^0}{w_{\text{eq}}^0} \frac{\partial w_{\text{eq}}^0}{\partial \alpha^\eta} + \frac{w_s^0 w_s^f - w_n^0 w_n^f}{w_{\text{eq}}^0} \quad (62)$$

$$\frac{\partial \alpha^\eta}{\partial \alpha} = \eta \alpha^{\eta-1} \quad (63)$$

$$\frac{\partial \alpha}{\partial \mathbf{w}_{ns}} = 2\alpha(1-\alpha) \begin{bmatrix} -1/\langle w_n \rangle \\ 1/w_s \end{bmatrix}. \quad (64)$$

### 3.3. Xu–Needleman cohesive law in local 2D coordinate frame

The second cohesive law that is used in the numerical examples is the one proposed by Xu and Needleman [44]. The cohesive law has been widely used for various crack growth analysis: for brittle solids [44,45], particle/matrix interfaces [46], elastic-viscoplastic solids [47], steel-PMMA bimaterial interfaces [48], alumina/titanium composites [49], rough interface between two solids [50], elastic-plastic graded materials [51], strictly for concrete [14,52–55] and many others.

Other kinds of cohesion law applied for concrete can be found in literature where some of them have initial stiffness while others are ridged, e.g. [21,40,56,57,33,58]. A nice description of some traction-separation relationships can be found in [59].

The formulas for the normal and shear cohesive tractions are as follows:

$$t_n = e F_t \frac{\kappa}{\delta_n} \exp\left(-\frac{\kappa}{\delta_n} - \frac{w_s^2}{\delta_s^2}\right), \quad (65)$$

$$t_s = \mu \frac{w_s}{\delta_s}, \quad (66)$$

where  $e = \exp(1)$ ,  $\delta_n$  and  $\delta_s$  are model parameters, and  $\mu$  is the shear modulus of the material,  $\kappa$  relates to normal separation and reads:  $\kappa = \xi \delta_n + w_n$ . It means a linear coordinate shifting about  $\delta_n$  in the traction-separation diagram. Taking  $\xi = 0$  the model returns to the standard Xu–Needleman model and taking  $\xi = 1$  the model comes to the brittle failure [54]. The parameter  $\delta_n$  can be set from the relation:

$$G_F = \int_0^\infty t_n(w_n, w_s = 0) dw_n, \quad (67)$$

which implies

$$\delta_n = \frac{G_F}{F_t \exp(1-\xi) \cdot (\xi + 1)}. \quad (68)$$

The elements of the cohesive stiffness matrix  $\mathbf{C}_{ns}$  have the form:

$$\frac{\partial t_n}{\partial w_n} = e F_t \exp \left( -\frac{\kappa}{\delta_n} - \frac{w_s^2}{\delta_s^2} \right) \left( \frac{1}{\delta_n} - \frac{\kappa}{\delta_n^2} \right), \quad (69)$$

$$\frac{\partial t_n}{\partial w_s} = -2 e F_t \frac{\kappa}{\delta_n} \frac{w_s}{\delta_s^2} \exp \left( -\frac{\kappa}{\delta_n} - \frac{w_s^2}{\delta_s^2} \right), \quad (70)$$

$$\frac{\partial t_s}{\partial w_n} = 0, \quad (71)$$

$$\frac{\partial t_s}{\partial w_s} = \frac{\mu}{\delta_s}. \quad (72)$$

### 3.4. Bilinear and exponential cohesive laws in local 2D coordinate frame

Bilinear and exponential cohesive laws are widely used for modeling cohesive cracks in quasi-brittle materials like concrete, e.g. [60,61]. The bilinear softening law is characterised by four parameters: (i) tensile strength  $F_t$ , (ii) fracture energy  $G_F$  corresponding to the area under the entire cohesive law, (iii) the initial fracture energy  $G_f$  corresponding to the area under the initial tangent of the law, and iv) the kink stress  $F_k$ . Additionally the initial stiffness often applied what requires additional parameter called here  $w_i$ :

$$t_n = \begin{cases} F_t \cdot w_n / w_i & \text{for } w_n < w_i \\ F_t \cdot (1 - (w_n - w_i) / (w_f - w_i)) & \text{for } w_i < w_n \leq w_k \\ F_k \cdot (1 - (w_n - w_k) / (w_c - w_k)) & \text{for } w_k < w_n \leq w_c \\ 0 & \text{for } w_c < w_n \end{cases} \quad (73)$$

where  $w_f = 2G_f / F_t$ ,  $w_k = w_f - F_k / F_t \cdot (w_f - w_i)$ ,  $w_c = 2(G_f - G_f) / F_k + w_f$ . In the situation when  $G_f = G_F$  the cohesion law reduces to standard triangular cohesion law.

The elements of the cohesive stiffness matrix  $\mathbf{C}_{ns}$  have the form:

$$\frac{\partial t_n}{\partial w_n} = \begin{cases} F_t / w_i & \text{for } w_n < w_i \\ -F_t / (w_f - w_i) & \text{for } w_i < w_n \leq w_k \\ -F_k / (w_c - w_k) & \text{for } w_k < w_n \leq w_c \\ 0 & \text{for } w_c < w_n \end{cases} \quad (74)$$

The exponential softening law has the following form:

$$t_n = F_t \exp \left( -\frac{F_t}{G_F} w_n \right) \quad (75)$$

The elements of the cohesive stiffness matrix  $\mathbf{C}_{ns}$  for the exponential law have the form:

$$\frac{\partial t_n}{\partial w_n} = -\frac{F_t^2}{G_F} \exp \left( -\frac{F_t}{G_F} w_n \right) \quad (76)$$

The shear cohesive tractions is taken the same as in Section 3.3.

## 4. Numerical examples

The proposed formulation has been implemented and tested in a three dimensional XFEM model. Three examples have been selected to test the formulation and implementation. The first two cases are delamination cases in which the cracks are flat but the direction of the traction vector changes considerably over the cohesive zone. The third case, which concerns cracking in concrete, has a curved crack plane.

The delamination examples are analyzed with Turon's cohesive law. In order to mimic initially rigid behavior, the cohesive law is shifted such that the initial traction for zero crack opening is equal

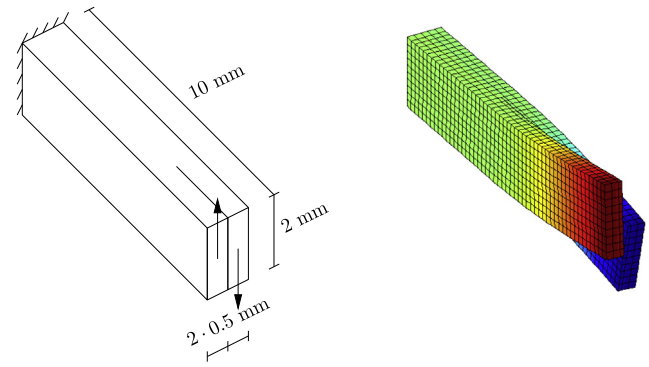


Fig. 1. Test setup and deformation.

Table 1

Material parameters for numerical examples.

|                            |           |                                 |
|----------------------------|-----------|---------------------------------|
| Young's modulus            | $E$       | $100 \cdot 10^3 \text{ N/mm}^2$ |
| Poisson ratio              | $\nu$     | 0.3                             |
| Tensile strength           | $F_t$     | $50 \text{ N/mm}^2$             |
| Shear strength             | $F_{sh}$  | $100 \text{ N/mm}^2$            |
| Mode I fracture toughness  | $G_{Ic}$  | $0.2 \text{ N/mm}$              |
| Mode II fracture toughness | $G_{IIc}$ | $0.5 \text{ N/mm}$              |
| Mode interaction parameter | $\eta$    | 2                               |
| Interface dummy stiffness  | $K$       | $10^6 \text{ N/mm}^3$           |

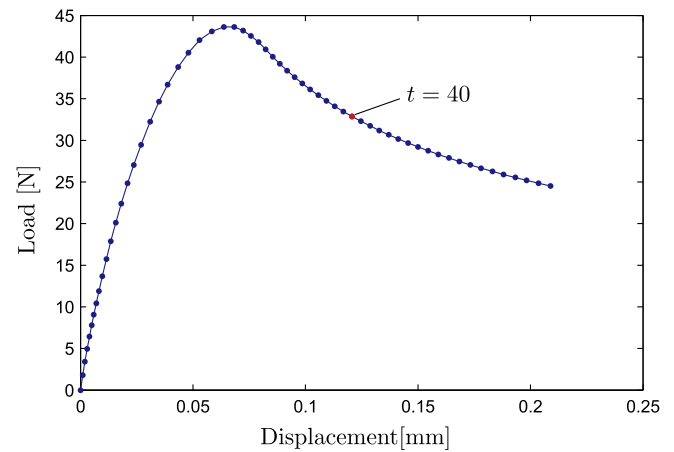


Fig. 2. Load-displacement relation for tearing test; the highlighted time step corresponds to the deformed mesh in Fig. 1 and the tractions in Fig. 3.

Table 2

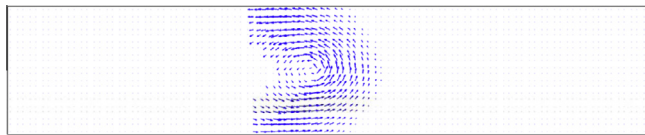
Convergence of Newton-Raphson procedure for time step  $t = 40$  in tearing test.

| Iteration number | Relative residual norm |
|------------------|------------------------|
| 1                | 3.8505e-02             |
| 2                | 5.1293e-03             |
| 3                | 5.6438e-05             |
| 4                | 3.5676e-09             |
| Crack growth     |                        |
| 1                | 1.3395e-04             |
| 2                | 2.8702e-09             |

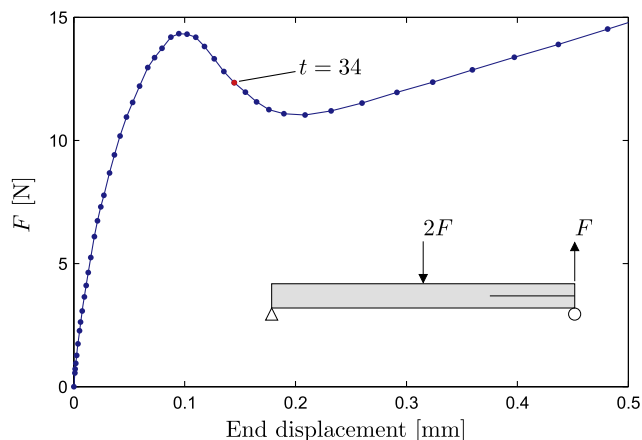
to the traction on the crack surface as computed from the stress matrix at the moment at which the crack segment is introduced. This concept has first been introduced by Hille et al. [62] and applied to Turon's cohesive law by Van der Meer et al. [41]. In

the current context, the shift must be applied to the full displacement jump  $[[\mathbf{u}]]$  rather than to  $w_n$  and  $w_s$ .

The cracks in the delamination examples are flat, which in combination with regular meshes means that complications that exist for crack propagation algorithms in three dimensions can be avoided. The employed crack propagation algorithm works as follows: when all cohesive integration points in a given cracked



**Fig. 3.** Tractions from a single time step in the tearing test. Arrows indicate magnitude and direction of traction vectors in every integration point in the interface that forms the fracture plane (crack grows from right to left).



**Fig. 4.** Mixed-mode bending test.

**Table 3**

Convergence of Newton–Raphson procedure for time step  $t = 34$  in mixed-mode bending test.

| Iteration number | Relative residual norm |
|------------------|------------------------|
| 1                | 3.8140e–01             |
| 2                | 1.1738e–01             |
| 3                | 6.0922e–03             |
| 4                | 2.0720e–05             |
| 5                | 2.4196e–10             |
| Crack growth     |                        |
| 1                | 2.2123e–03             |
| 2                | 7.3598e–07             |

element have entered the softening regime, it is ensured that this element is kinematically completely free to open, by enriching all nodes of this element (and hence inserting crack segments in all neighboring elements). In newly cracked elements the strength has typically not yet been reached, such that they are initiated in the hardening regime.

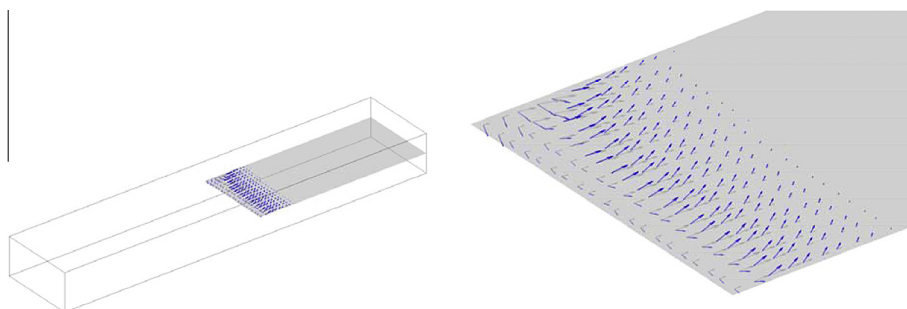
The first case is a tearing test (see Fig. 1). In this test the displacement jump is a pure sliding displacement, but the direction of the sliding varies strongly. The case is analyzed with an initial notch of 2 mm and the material parameters from Table 1. An arc-length solver with adaptive time stepping [63] is used. The load versus crack opening response that has been obtained with the proposed formulation is visualized in Fig. 2. For each time step a dot is included in the graph; it can be observed that the increments are fairly large for a cohesive crack growth analysis. The time step  $t = 40$  is indicated which corresponds to the other results shown for this example. In Table 2, the convergence of the Newton–Raphson procedure is demonstrated for this time step with the values for the scaled residual from each iteration. Note that the iterative procedure converges twice for this time step, once before crack growth and once after crack growth.

A typical traction distribution from this example is shown in Fig. 3. As the cohesive zone moves from right to left, the direction of  $\mathbf{t}$  and hence of  $\mathbf{s}_d$  changes over time in the integration points. Nevertheless, quadratic convergence is obtained in the simulations. This indicates that the linearization is complete, also with respect to changes in  $\mathbf{s}_d$ .

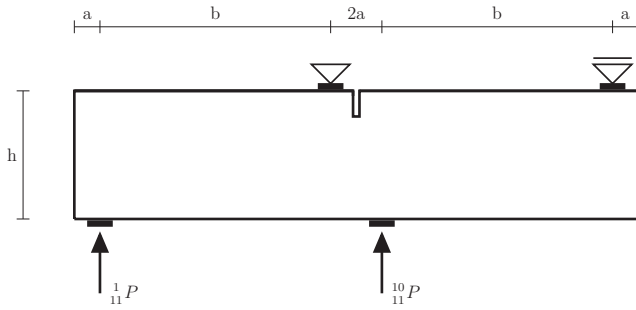
The second test is a mixed-mode bending test (see Fig. 4). Here, the performance of the formulation is tested for cases where the normal and shear contributions vary over the cohesive zone. The model geometry and parameters are the same as in the previous example although symmetry boundary conditions are applied so that the model can be understood as half of a 4 mm wide beam.

Also in this test, convergence is generally good except for the first time step in which the contact zone develops that exists where the crack front meets the free edge. The evolution of the relative residual for a representative time step is reported in Table 3. Again, quadratic convergence is obtained, which indicates that the cohesive law behaves regularly in this mixed-mode case and that the linearization is exact (see Fig. 5).

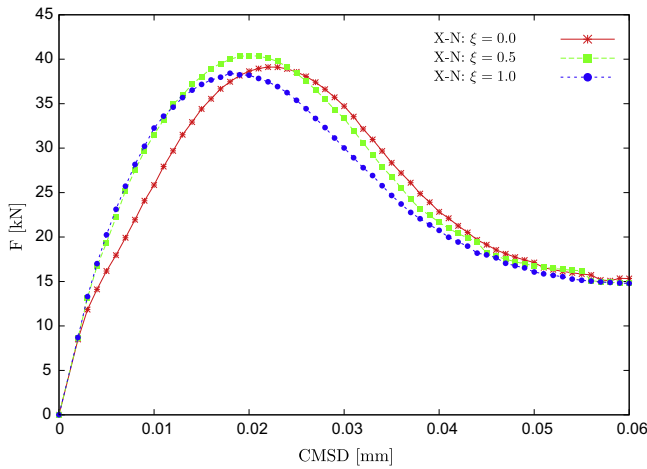
As a third example, the four-point-shear loading of the single-edge notched concrete beam is examined. The beam has been experimentally and numerically analysed by Schlangen [64]. The geometry with boundary conditions of the specimen are shown in Fig. 6. In the experiment a curved crack path is observed due to antisymmetric beam loading. The crack goes from the right corner of the notch to the lower right loading platen. For numerical three-dimensional analysis the domain has been discretized with 2930 tetrahedral finite elements of second order. In this example, three different cohesive laws are applied: the Xu–Needleman law as well as bilinear and exponential cohesive laws. The used



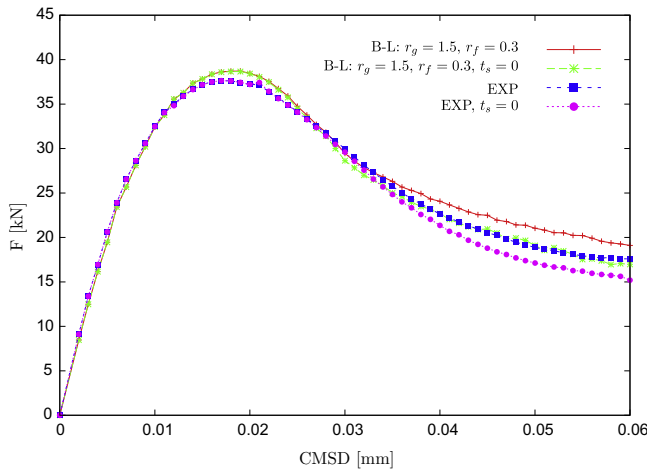
**Fig. 5.** Tractions from a single time step in the mixed-mode bending test: overview of crack location (left) and closeup of spatial distribution of tractions (right).



**Fig. 6.** Geometry and boundary condition of four-point-shear loading SEN beam with:  $a = 20$  mm,  $b = 180$  mm,  $h = 100$  mm, notch length = 20 mm, notch width = 5 mm, beam thickness = 100 mm, loading platens = 20 mm.

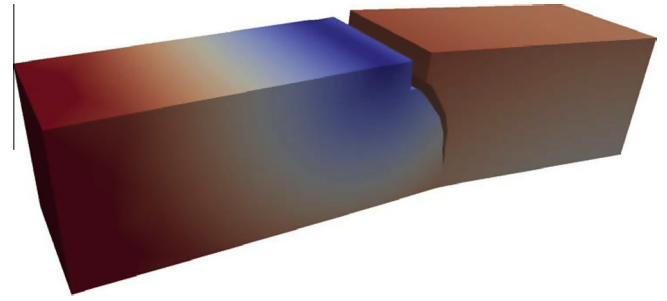


**Fig. 7.** The equilibrium paths in load versus CMSD for cracking of the single-edge notched concrete beam using Xu-Needleman (X-N) cohesive crack model for various  $\xi$  parameters.



**Fig. 8.** The equilibrium paths in load versus CMSD for cracking of the single-edge notched concrete beam bilinear (B-L) and exponential (EXP) cohesive crack models.

material parameters are (cf. [64,65]): Young's modulus  $E = 35$  GPa, Poisson's ratio  $\nu = 0.2$ , tensile strength  $f_t = 3$  MPa, and fracture energy  $G_F = 0.08$  N/mm. Following [54] the cohesive shear stiffness  $\mu/\delta_s$  is set to 10 [N/mm]. Additionally for the bilinear model additional parameters are defined: a fracture energy ratio  $r_g = G_F/G_f$  and a stress ratio  $r_f = F_k/F_t$ .



**Fig. 9.** The single-edge notched concrete beam in advanced process stage of cracking.

Following literature on this setup, the results are shown in the form of the relation between the load and the sliding part of the crack opening vector or crack mouth sliding displacement (CMSD). Simulation results are presented in Figs. 7 and 8 where the cases with active and dis-active shear part of cohesion tractions are considered. The results obtained in the example follow the conclusions of other researchers that the damage in isotropic materials is dominated by normal part of cohesion tractions, e.g. in [13,66,67]. In Fig. 9 the cracked beam is shown in the advanced stage, which similar for all analysed cases. The results are in good agreement with the experimentally observed crack evolution path and loading-CMSD relation [64,65].

## 5. Conclusion

In this paper a consistent iterative scheme for cohesive crack analysis using XFEM has been presented. Firstly the mathematical model has been presented which assumes inner displacement discontinuity with cohesion forces along the crack surface. The mathematical model is presented in global formulation in iterative scheme, where nonlinearity connected with cohesion forces is assumed. The cohesion forces generally depend on the crack opening. Therefore a cohesion stiffness matrix is needed in the iterative mathematical model.

To find solution of the mathematical model an approximation of displacement field has to be assumed as well as the approximation of discontinuity along crack surface. In this paper, an XFEM approximation has been chosen, where special discontinuous functions are included in the solution. The system of equations that follows from the discretization is solved iteratively with the Newton-Raphson method, which requires evaluation of the tangent stiffness matrix. In the paper a new approach for calculating the part of this stiffness matrix that is related to the cohesion forces has been presented. The presented method is valid for both 2D and 3D cases and it is formulated such that the cohesive law can always be evaluated in only two dimensions.

The proposed formulation has been implemented and verified with numerical examples of delamination and of cracking in concrete. The presented algorithm for calculating the cohesion stiffness matrix is effective and has a general form. The algorithm in the paper was applied for cohesion cracks, but it is also suitable for contact problems, particularly with a penalty approach as present in many existing cohesive laws.

An open question is how to deal with the situation when the direction of the sliding traction depends on the complete history of the sliding crack opening vector  $\mathbf{w}_s$  as is the case with Coulomb's law of friction. This is a topic for further research.

## Acknowledgment

This scientific research has been carried out as a part of the project "Innovative recourses and effective methods of safety

improvement and durability of buildings and transport infrastructure in the sustainable development” financed by the European Union from the European Fund of Regional Development based on the Operational Program of the Innovative Economy.

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